

1. The purpose of a loop structure is to repeatedly execute a block of code while a specified condition is true.
2. A while loop checks its condition before executing the loop body, so it may execute zero times. A do-while loop executes the loop body first and checks the condition afterward, so it executes at least once.
3. The input validation loop was used in the programs where user input had to be checked to make sure it was within an acceptable range before continuing.
4. a) An infinite loop is a loop that continues executing forever because its condition never becomes false.

b) Two errors that can cause an infinite loop:

- The loop control variable is never changed.
- The loop control variable is changed in the wrong direction.

c) Overflow occurs when a calculation produces a value larger than the maximum value that can be stored in the data type.

5. The loop executes 60 times
6. Any initial value of x less than 120 would make the loop infinite.
7. Counters:
 - Keep track of how many times something occurs.
 - Count the number of items, repetitions, or events.

Accumulators:

- Keep a running total.
- Add values together, such as calculating a sum or total cost.

8. `int sum = 0;`

```
for (int x = 3; x <= 10; x++) {  
    sum += x;  
}
```

The final value of sum is 52.

9. Two factors to consider when choosing a loop structure are:
 - Whether the number of repetitions is known.
 - Whether the loop needs to execute at least once before checking the condition.
10. a) Two methods for debugging an application:

- Use a debugger/breakpoints.
- Add output statements to display variable values.

b) Using a debugger would be best because the random numbers can be examined while the program runs, allowing you to see exactly what values are being generated and how they affect the if statement.

11. a) 9

b) "my "

c) "my string"

d) "MY STRING"

e) "my string"

12. a) false

b) true

c) -8

d) 0

e) -8

f) 9

g) 9

h) true

i) true

13. a) Get the sentence

Set wordCount to 0

Split the sentence into words

For each word:

 Increase wordCount by 1

Display wordCount

b) Get the sentence

Set letterCount to 0

For each character in the sentence:

 If the character is a letter:

 Increase letterCount by 1

Display letterCount

14. a) False: A while statement checks the condition before executing the loop

b) True

c) False: An accumulator keeps a running total; it does not indicate when a loop should stop

d) False: A sentinel can be any predetermined value used to signal the end of input.

e) False: A variable declared in a for statement generally has scope only within that for loop.

- f) False: String is a class/reference type, not a primitive data type.
- g) False: String indexing starts at position 0.
- h) False: Strings should be compared using methods such as `.equals()`, not relational operators such as `>`.
- i) True